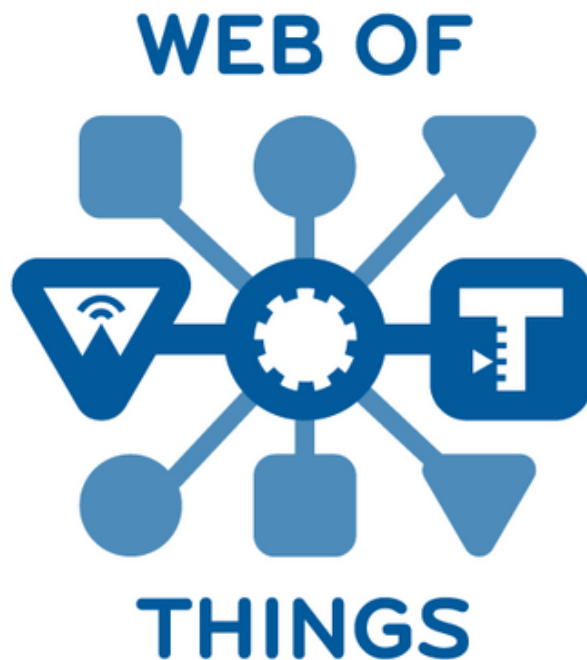


Cloud of Things Scope Statement

Smart Greenhouse

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Chapter 1

Concept

The concept of this project is to design and implement a **Smart Greenhouse System** based on Cloud of Things technologies, including the Internet of Things (IoT), cloud computing, and artificial intelligence (AI), to optimize plant growth conditions. The system will use sensors to continuously collect environmental data such as temperature, humidity, and water levels, as well as soil quality metrics (N, P, K). This information will be transmitted to the cloud for storage, processing, and analysis. The integration of an artificial intelligence component will allow for the automatic adjustment of actuators (fans, water pumps) to maintain an optimal environment, anticipate plant needs, and prevent unfavorable growth conditions.

1.1 Problematic

Traditional greenhouse management is often manual and imprecise, leading to inefficient water and energy consumption and vulnerability to environmental changes. It's difficult to maintain a consistent climate and soil quality, which can harm crop productivity. Furthermore, farmers often lack access to real-time data to make informed decisions. There is a pressing need for smart systems that can proactively monitor and control greenhouse environments, automating irrigation and ventilation processes while providing accurate information to farmers.

1.2 Context of the Project

This project is at the intersection of IoT, cloud computing, and AI. Recent advancements in sensors and mobile connectivity make it possible to capture agricultural data outside of lab environments. At the same time, cloud services provide the scalability and computational power necessary to process and analyze this data in real time. The integration of machine learning models opens the door to predicting growth conditions and triggering automated corrective actions. Within this technological context, our project demonstrates how the Cloud of Things can enable smarter, more sustainable agricultural solutions.

1.3 Ambitions

The ambitions of this project are to:

- Improve crop yield and quality by maintaining optimal environmental conditions.
- Automate irrigation and ventilation tasks to reduce workload and human error.
- Optimize water and energy usage through data-driven controls.
- Provide farmers with real-time data insights via a dashboard.
- Explore the potential of machine learning in agriculture by predicting plant water and nutrient needs.

Chapter 2

Target Audience

The Smart Greenhouse System is intended for two main groups:

- **Commercial farmers and horticulturists:** Who require a reliable solution to optimize growing conditions and maximize their yield.
- **Gardening enthusiasts and individuals:** Who want to grow plants at home with an automated system, ensuring their health and well-being.

Chapter 3

Equipment

The following equipment list is temporary and may evolve as the project progresses. It represents the initial hardware components considered for implementing the Smart Greenhouse System.

Equipment	Description
Raspberry Pi 4/5	Serves as the central processing and communication unit, handling sensor data aggregation and cloud connectivity.
DHT11/DHT22 or SHT31 Sensors	Measure the temperature and relative humidity of the greenhouse air.
Water Level Sensor	Detects the water level in a reservoir for irrigation (ultrasonic, capacitive, or analog float).
NPK Soil Quality Sensor	Measures the levels of Nitrogen (N), Phosphorus (P), and Potassium (K) in the soil.
Relay	Enables the control of actuators (fan, pumps) by switching the electrical supply.
Electric Fan	Used for ventilation and temperature control.
Water Pump	Actuates plant irrigation (pump for localized irrigation and a main pump).

Chapter 4

Functionalities

4.1 Sensor Data Acquisition

The system will continuously collect environmental and soil data using the integrated sensors:

- Temperature and humidity via the DHT or SHT sensor.
- Water level via the appropriate sensor.
- Nitrogen, Phosphorus, and Potassium (NPK) levels with the soil quality sensor.

Raw data will be processed locally on the Raspberry Pi for efficiency and securely transmitted to the cloud backend for storage and analysis.

4.2 Mobile Application Development

A mobile application, developed as a Progressive Web App (PWA) and Single Page Application (SPA), will serve as the main user dashboard. The app will provide the following functionalities:

- Real-time visualization of sensor data (temperature, humidity, water level, NPK).
- Manual control of actuators (fan, pumps).
- Notifications and alerts if predefined thresholds are exceeded (e.g., temperature too high or low water level).
- A user-friendly interface accessible from various devices without requiring installation.

4.3 Backend and Communication

The backend will be developed using **Jakarta EE** as the middleware, providing a robust and scalable platform for processing and managing the data. Communication between the IoT devices and the backend will be handled by an **MQTT broker**, which is an efficient and lightweight messaging protocol ideal for IoT environments.

4.4 Security

The security of the system will be a top priority. **Identity and Access Management (IAM)** will be used to manage and secure access to the cloud resources and data. This includes authenticating devices and users, and authorizing their access to specific functionalities or data streams, ensuring patient privacy and compliance with health data regulations.

Chapter 5

Machine Learning Integration

5.1 Model Development

The goal is to develop a compact and efficient machine learning model to analyze sensor data and make automated decisions. Depending on the chosen approach, the model will be used for:

- **Irrigation Needs Prediction:** By using sensor data (soil moisture, temperature, etc.), the model can predict the plants' water needs and trigger the water pump.
- **Predictive Climate Control:** The model can anticipate temperature or humidity fluctuations and activate the fan or other control systems to maintain stable conditions.

5.2 MLOps (Machine Learning Operations)

An MLOps pipeline will be implemented to streamline the development, deployment, and monitoring of the predictive models. Its key features will include:

- **Automated Training and Deployment:** The MLOps framework will train and update the model with new sensor data to ensure the accuracy of real-time predictions.
- **Performance Monitoring:** Continuous monitoring will track the model's effectiveness to detect drift and maintain its reliability.
- **Alerts and Notifications:** In case of critical conditions (e.g., risk of drought or overheating), the system will trigger immediate alerts for the user.
- **Data Management:** Secure data tracking and versioning will support model improvement.
- **Optimization and Feedback:** The model will be continuously refined based on real-world usage to optimize performance on local hardware.

5.3 Dataset

For model training, we will use the public dataset "IoT Agriculture 2024" available on Kaggle: [Link](#).

This dataset contains variables such as temperature, humidity, and soil quality data that are relevant to our project. We will explore two modeling approaches:

- **Prophet:** A model developed by Facebook, suitable for time-series data, to predict water needs or temperature trends.
- **LSTM (Long Short-Term Memory):** A type of recurrent neural network (RNN) capable of capturing long-term dependencies in time-series data, which could offer more accurate predictions.